
Department of Civil Engineering, IIT Madras
CV2090 (Probability & Statistics for Civil Engineers)
Jul–Nov 2026

7 Sep, 2026

Tutorial-4

3:00–5:00 pm

Instructions:

1. Make assumptions wherever necessary, and *state* them explicitly.
 2. Tutorials are closed-phone, closed-tablet, and closed-LLM.
 3. You can bring books and notes, you are welcome to discuss with friends and TAs. However, your submission should be your own work.
 4. Submit handwritten solutions on A4 sheets, prepared during the tutorial hours.
 5. Submissions will not be accepted after the tutorial hours.
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Q1. Joint Discrete Random Variables Two discrete random variables X and Y have the joint probability mass function $p_{XY}(x, y)$ shown below:

$p_{XY}(x, y)$	$y = 1$	$y = 2$
$x = 1$	2/12	3/12
$x = 2$	3/12	4/12

- (a) Find the marginal PMFs $p_X(x)$ and $p_Y(y)$.
- (b) Determine whether X and Y are statistically independent.
- (c) Calculate the expected values $\mathbb{E}[X]$, $\mathbb{E}[Y]$, and $\mathbb{E}[XY]$.
- (d) Find the covariance $\text{COV}(X, Y)$, standard deviations σ_X and σ_Y , and the correlation coefficient ρ .

Q2. Joint Continuous Distributions Let X and Y be two continuous random variables representing normalized structural load components with marginal cumulative distribution functions $F_X(x) = 1 - e^{-x}$ for $x \geq 0$ and $F_Y(y) = y$ for $0 \leq y \leq 1$. Suppose their joint cumulative distribution function is given by the product of their marginal CDFs, such that $F_{XY}(x, y) = F_X(x)F_Y(y)$.

- (a) Write down the explicit expression for the joint CDF $F_{XY}(x, y)$ in terms of x and y .
- (b) Derive the joint probability density function $f_{XY}(x, y)$ by evaluating the mixed partial derivative:

$$f_{XY}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x, y).$$

- (c) Evaluate the joint probability $P(X \leq 1, Y \leq 0.5)$ directly from the joint CDF expression.

Note on Copulas and Sklar's Theorem: A copula is a function that links univariate marginal distributions to form a joint distribution. By Sklar's Theorem, any joint cumulative distribution function $F_{XY}(x, y)$ can be decomposed into its marginal distributions and a copula $C(u, v)$, allowing engineers to model marginal behaviors and dependency structures separately.

Q3. Functions of Multiple Random Variables and Covariance Propagation

In engineering design, structural demand often consists of combined load effects. Let D (dead load) and L (live load) be two correlated random variables with means $\mu_D = 100$ kN, $\mu_L = 50$ kN, standard deviations $\sigma_D = 10$ kN, $\sigma_L = 15$ kN, and a correlation coefficient $\rho_{DL} = 0.4$. The total load is given by the linear combination $T = D + L$.

- (a) Using the properties of expectation and variance for linear combinations of random variables, derive the general formula for the variance $\text{Var}(T)$ in terms of $\text{Var}(D)$, $\text{Var}(L)$, and $\text{COV}(D, L)$.
- (b) Calculate the numerical mean μ_T and standard deviation σ_T of the total load T .
- (c) If the structural capacity is a deterministic limit $R = 140$ kN, estimate the probability of structural overload $P(T > R)$, assuming T follows a normal distribution and letting $\Phi(\cdot)$ represent the standard normal CDF.

Q4. Bivariate Normal Distribution Consider two random variables X and Y having the joint probability density function

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x-\mu_X}{\sigma_X} \right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X} \right) \left(\frac{y-\mu_Y}{\sigma_Y} \right) + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 \right\} \right]$$

where $-\infty < x, y < \infty$, $\sigma_X, \sigma_Y > 0$, and $-1 < \rho < 1$.

- (a) Obtain the marginal probability density function of X by integrating the joint PDF over y :

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy.$$

What does the marginal density function look like?

- (b) Using (a), write the marginal PDF of Y .
- (c) Find the conditional PDF $f_{Y|X}(y|x)$ using $f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}$.

By completing the square, show that the conditional distribution is

$$Y | X = x \sim N \left(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X), \sigma_Y^2 (1 - \rho^2) \right).$$

- (d) Interpret the role of ρ . In particular, what happens to the conditional mean and conditional variance when ρ is (i) 0, (ii) close to 1, and (iii) close to -1 ?

Q5. Optional Problem: Joint Continuous Distributions & Wedge-Domain Calculus Let X and Y be two continuous random variables representing structural capacity and demand whose joint probability density function is defined over a triangular wedge domain:

$$f_{XY}(x, y) = kxe^{-x}e^{-y/x} \quad \text{for } 0 < y < x < \infty$$

- (a) **Normalization:** Determine k such that $f_{XY}(x, y)$ integrates to 1 over the domain.
- (b) **Marginalization:** Derive the marginal PDF $f_X(x)$.
- (c) **Conditional Analysis:** Find the conditional PDF $f_{Y|X}(y|x)$ and evaluate the conditional expectation $\mathbb{E}[Y | X = x]$.
- (d) **Non-Linear Transformation:** Let $Z = \frac{Y}{X}$ represent the load-to-capacity demand ratio. Using the transformation (Jacobian) or double-integral probability mapping over the wedge domain, derive the exact PDF $f_Z(z)$ for $0 < z < 1$.